

Lesson 13 Demo 01

Creating Virtual Networking and Resolving Domain Names

Objective: To create Azure's virtual networking capabilities by creating a virtual network for Azure virtual machines, implementing network-based segmentation with static IP addresses, securing public endpoints, and enabling DNS name resolution within the virtual network

Tools Required: Azure account with administrator access

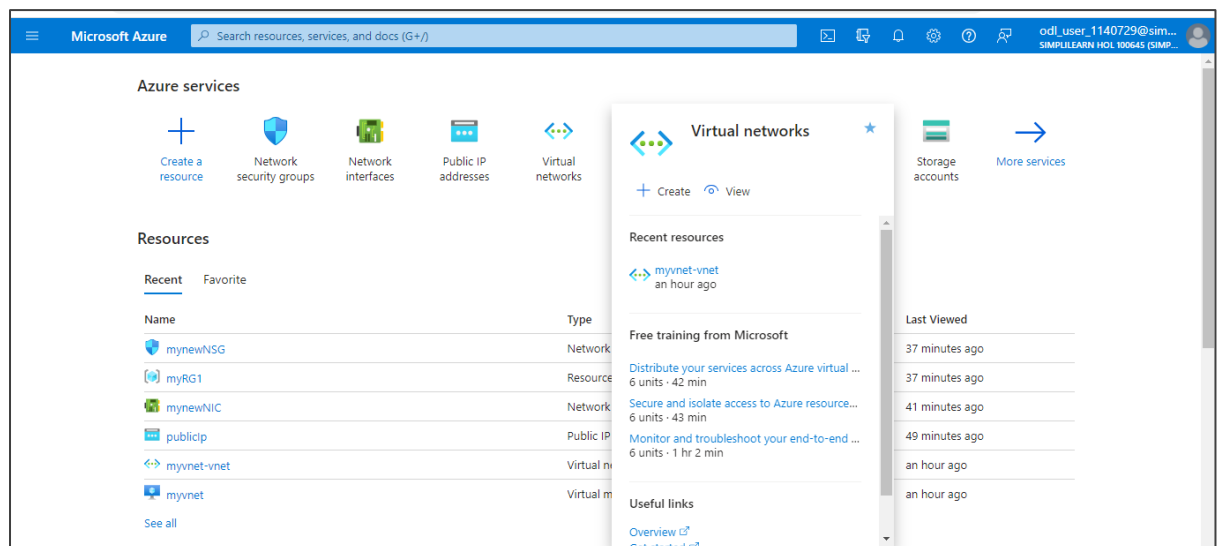
Prerequisites: None

Steps to be followed:

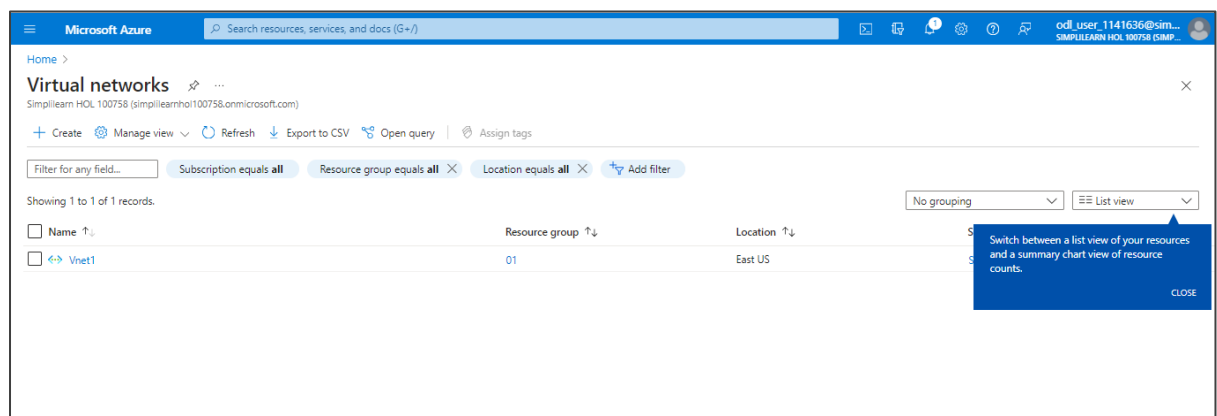
1. Create a virtual network
2. Create a subnet
3. Create a public IP address
4. Create the network interface
5. Create the network security group

Step 1: Create a virtual network

1.1 Log in to the Azure portal; search for and select **Virtual networks**



1.2 In the **Virtual networks** page, click on **+ Create**



1.3 In the **Create virtual network** page, navigate to the **Basics** tab and under **Resource group**, click on **Create new** to create a new resource group (**myRG2**)

Microsoft Azure Search resources, services, and docs (G+)

Home > Virtual networks >

Create virtual network

Basics Security IP addresses Tags Review + create

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Simplilearn HOL 100645

Resource group * myRG1 [Create new](#)

Instance details

Virtual network name *

Region (US) *

A resource group is a container that holds related resources for an Azure solution.

Name * myRG2

OK Cancel

1.4 Under **Instance details**, enter the **Virtual network name** as **myVnet1** and click on **Review + create**

Microsoft Azure Search resources, services, and docs (G+)

Home > Virtual networks >

Create virtual network

Basics Security IP addresses Tags Review + create

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Simplilearn HOL 100645

Resource group * (New) myRG2 [Create new](#)

Instance details

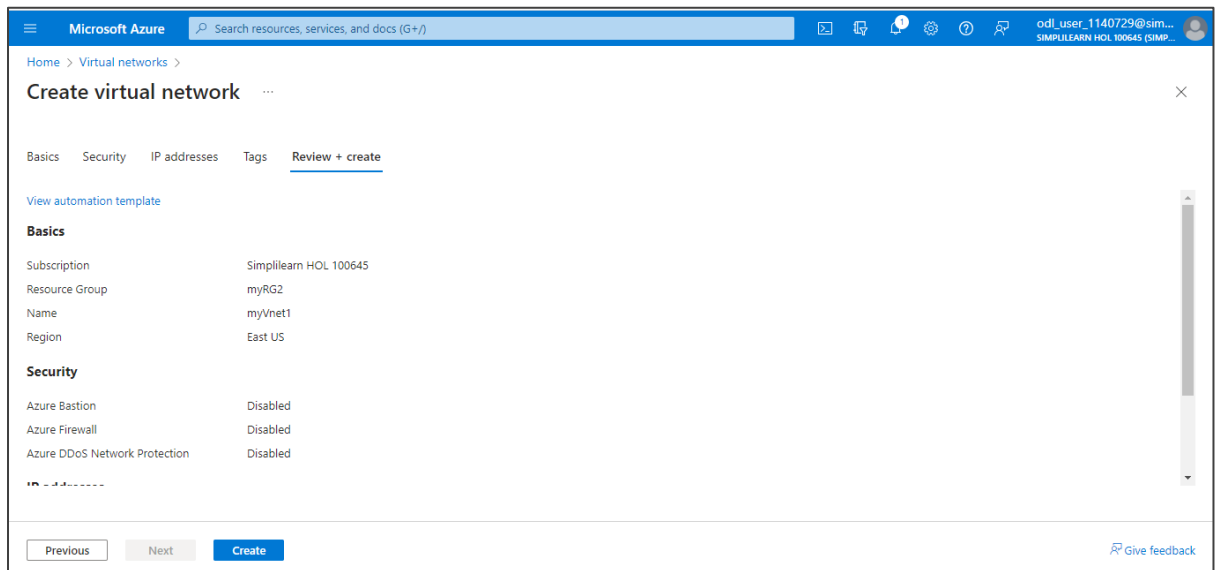
Virtual network name * myVnet1

Region (US) * (US) East US

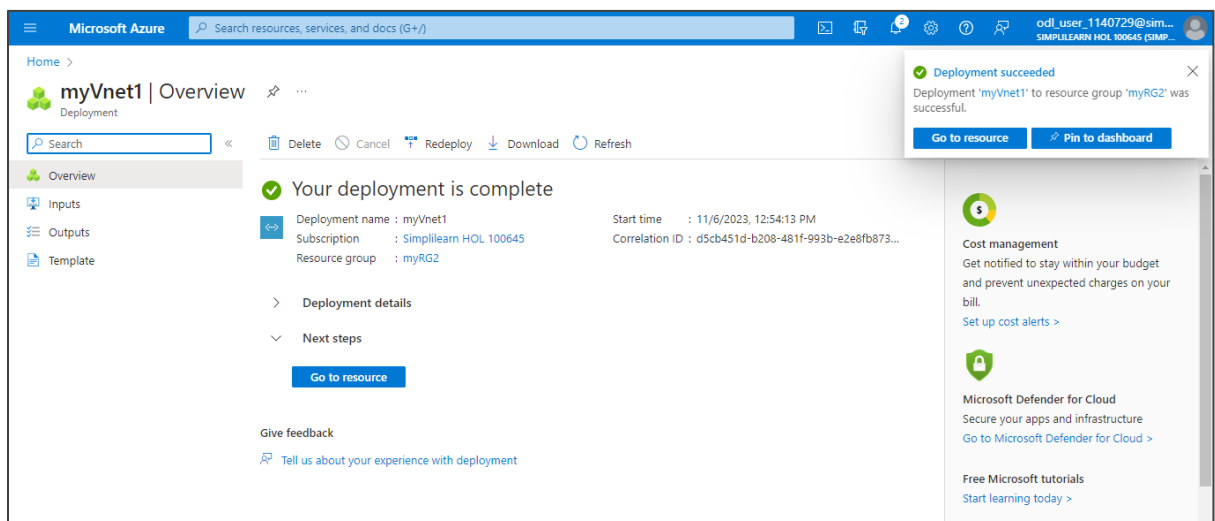
Previous Next **Review + create**

Give

1.5 Click on **Create**

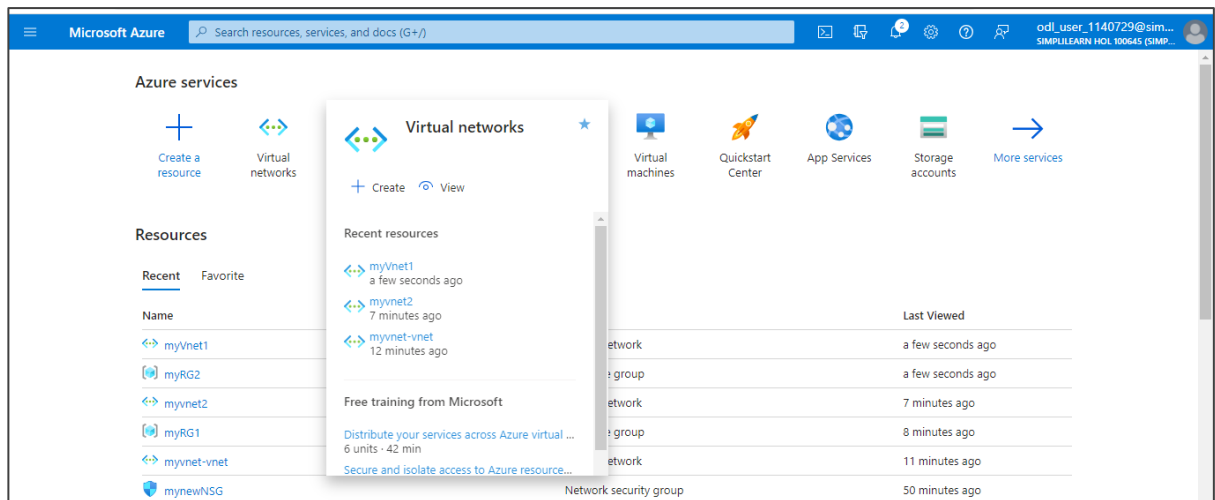


The output appears as shown below:

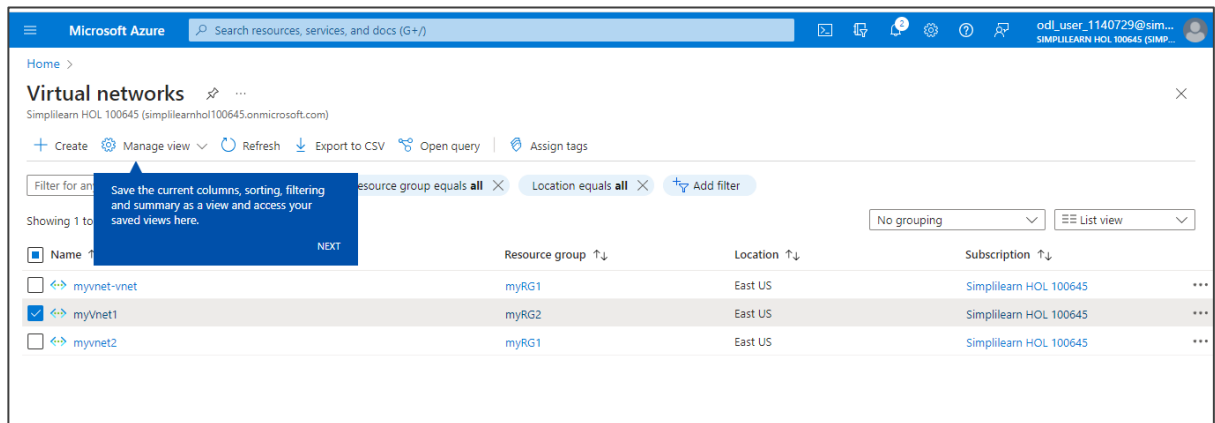


Step 2: Create a subnet

2.1 Navigate to Virtual networks



2.2 Choose the Virtual network (myVnet1) to which the user wants to add the subnet



2.3 Under **Settings**, select **Subnets**

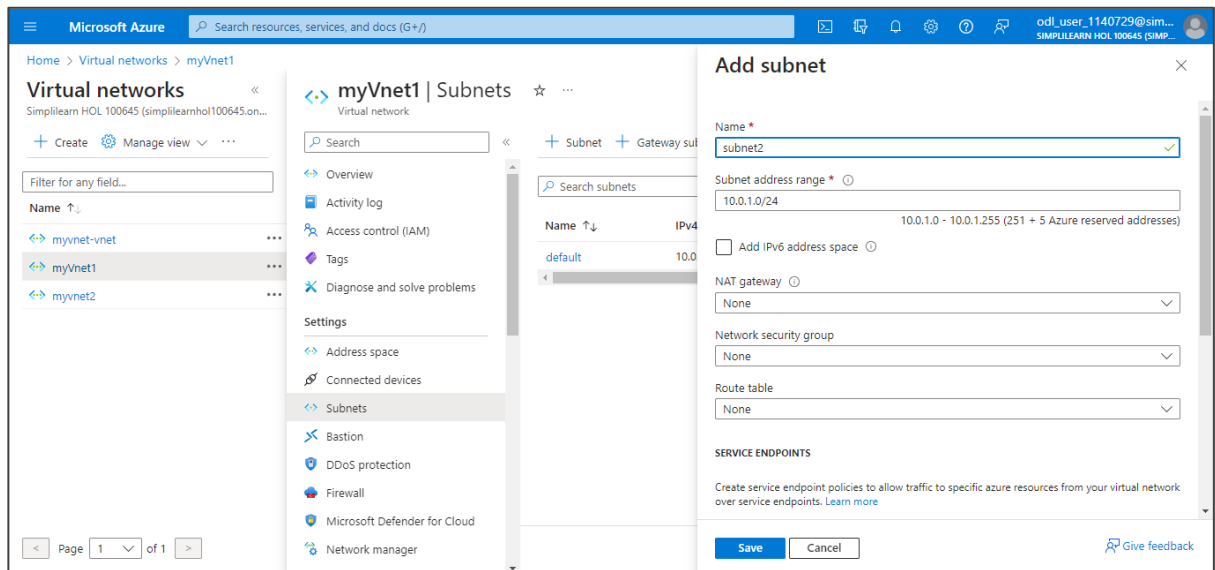
The screenshot shows the Microsoft Azure portal interface. On the left, the 'Virtual networks' list includes 'myVnet1'. The main pane displays the 'myVnet1' settings. In the left-hand navigation menu, the 'Subnets' option is highlighted with a red box. The right-hand pane shows the 'Essentials' section with various properties like Address space, DNS servers, and Subscription ID.

2.4 Click on **+ Subnet**

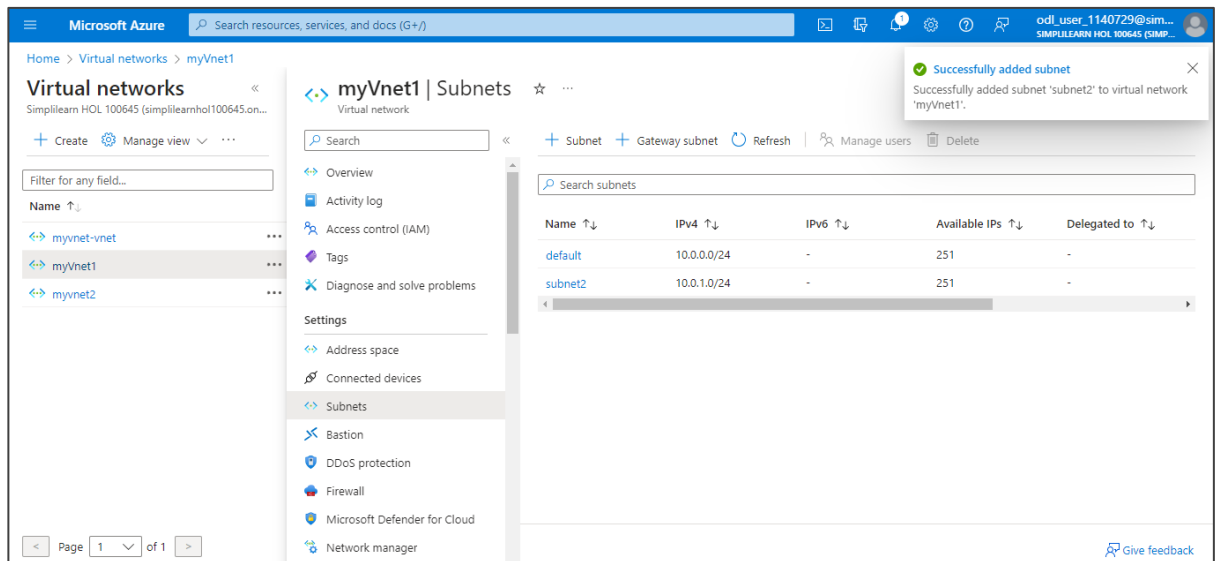
The screenshot shows the 'myVnet1 | Subnets' page in the Microsoft Azure portal. The '+ Subnet' button is visible at the top of the page. Below it, there is a table listing the subnets. The table has columns for Name, IPv4, IPv6, Available IPs, and Delegated to.

Name	IPv4	IPv6	Available IPs	Delegated to
default	10.0.0.0/24	-	251	-

2.5 In the **Add subnet** page, enter the **Name** as **subnet2** and click on **Save**

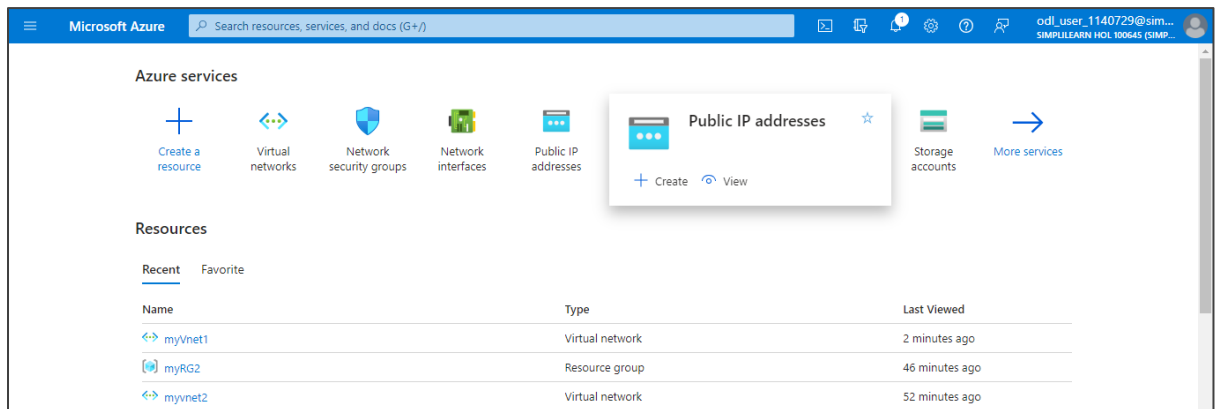


The output appears as shown below:

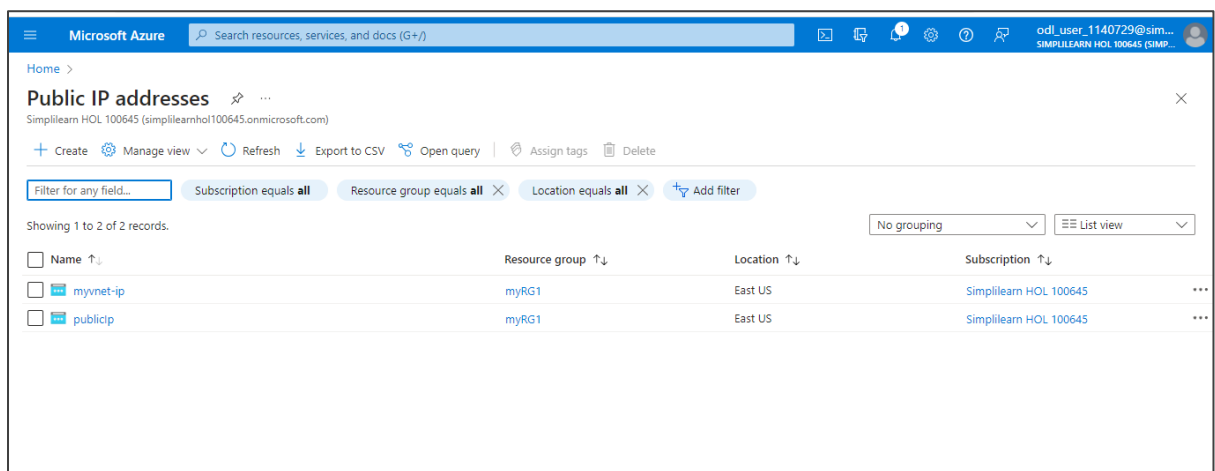


Step 3: Create a public IP address

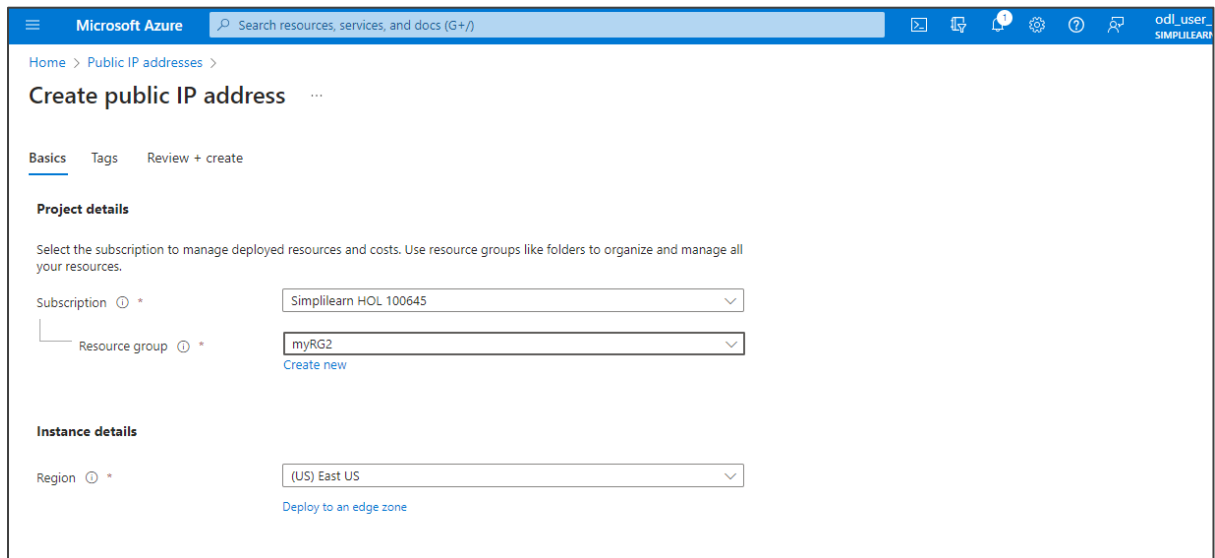
3.1 In the Azure portal, search for and select **Public IP addresses**



3.2 In the **Public IP addresses** page, click on **+ Create**



3.3 In the **Create public IP address** page, navigate to the **Basics** tab and enter the following details as shown in the screenshot below:



Microsoft Azure Search resources, services, and docs (G+)

Home > Public IP addresses >

Create public IP address

Basics Tags Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

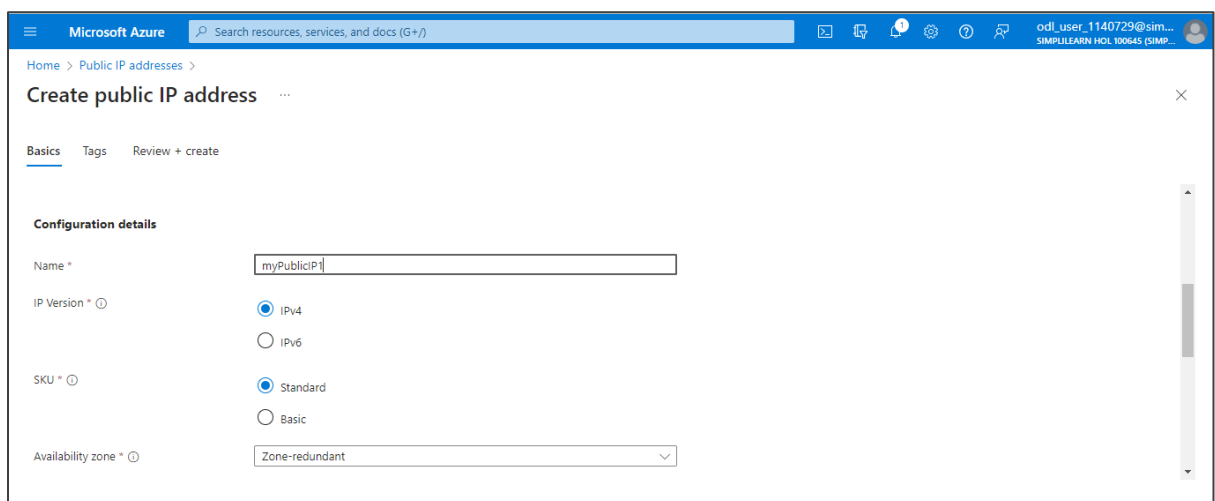
Subscription *

Resource group * [Create new](#)

Instance details

Region * [Deploy to an edge zone](#)

3.4 Under **Configuration details**, enter the **Name** as **myPublicIP1** and other details as shown in the screenshots below:



Microsoft Azure Search resources, services, and docs (G+)

Home > Public IP addresses >

Create public IP address

Basics Tags Review + create

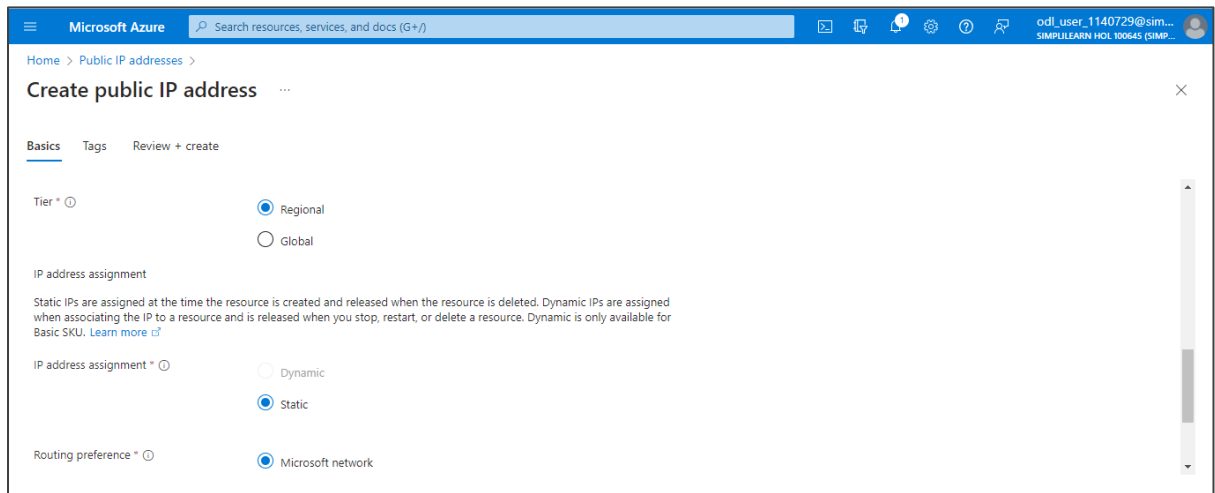
Configuration details

Name *

IP Version * ☒ IPv4 ☐ IPv6

SKU * ☒ Standard ☐ Basic

Availability zone *



Microsoft Azure Search resources, services, and docs (G+)

Home > Public IP addresses >

Create public IP address

Basics Tags Review + create

Tier * ☒ Regional ☐ Global

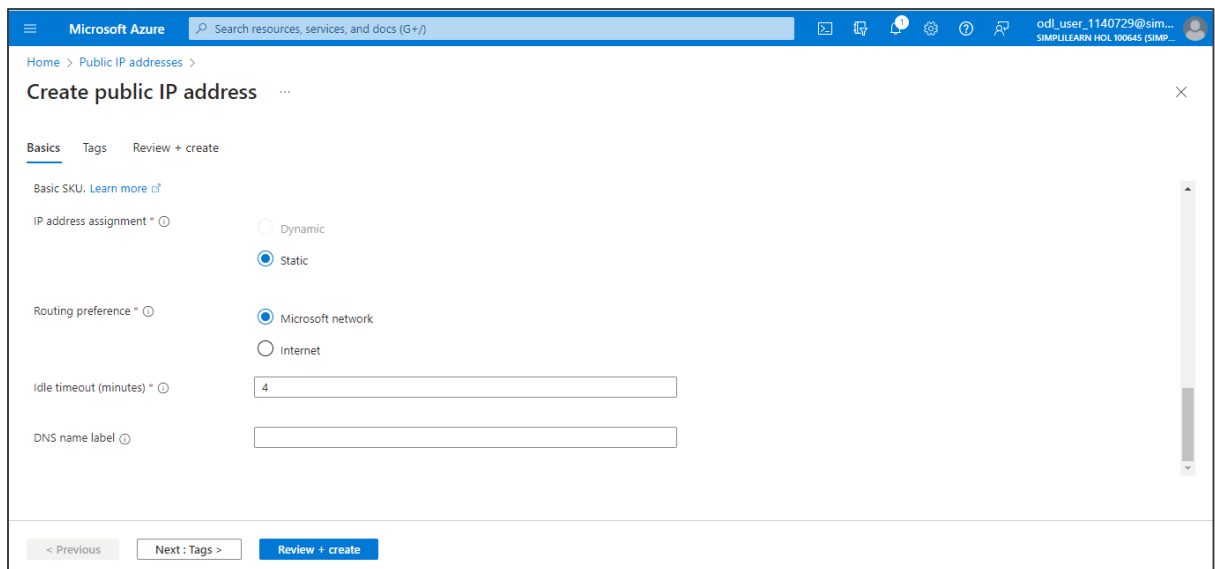
IP address assignment

Static IPs are assigned at the time the resource is created and released when the resource is deleted. Dynamic IPs are assigned when associating the IP to a resource and is released when you stop, restart, or delete a resource. Dynamic is only available for Basic SKU. [Learn more](#)

IP address assignment * ☐ Dynamic ☒ Static

Routing preference * ☒ Microsoft network

3.5 Click on **Review + create**



Microsoft Azure Search resources, services, and docs (G+)

Home > Public IP addresses >

Create public IP address

Basics Tags Review + create

Basic SKU. [Learn more](#)

IP address assignment * ☐ Dynamic ☒ Static

Routing preference * ☒ Microsoft network ☐ Internet

Idle timeout (minutes) *

DNS name label

< Previous Next : Tags > Review + create

3.6 After the validation passes, click on **Create**

Microsoft Azure Search resources, services, and docs (G+)

Home > Public IP addresses >

Create public IP address

Validation passed

Basics Tags Review + create

Basics

Subscription	Simplilearn HOL 100645
Resource group	myRG2
Region	eastus
Name	myPublicIP1
IP Version	IPv4
SKU	Standard
Tier	Regional
Availability zone	Zone-redundant
IP address assignment	Static
Routing preference	MicrosoftNetwork

< Previous Next > Create

The public IP address is created as shown below:

Microsoft Azure Search resources, services, and docs (G+)

Home >

PublicIPAddress-ARM | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

✓ Your deployment is complete

Deployment name : PublicIPAddress-ARM Start time : 11/6/2023, 1:42:34 PM
Subscription : Simplilearn HOL 100645 Correlation ID : 4972a8c8-1997-44b1-87b2-edfd38e21...
Resource group : myRG2

> Deployment details
v Next steps

Go to resource

Give feedback
Tell us about your experience with deployment

Deployment succeeded
Deployment 'PublicIPAddress-ARM' to resource group 'myRG2' was successful.
Go to resource Pin to dashboard

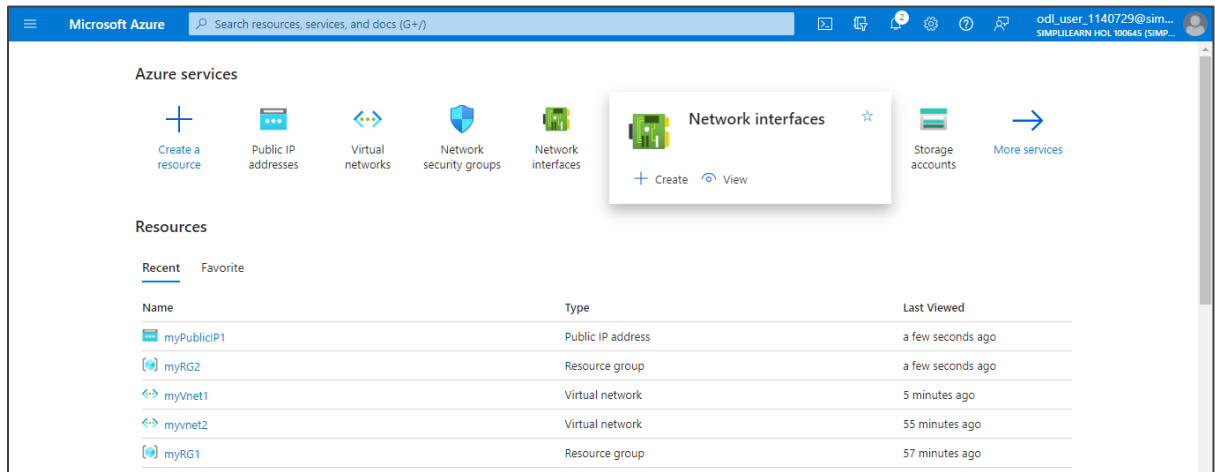
Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >

Microsoft Defender for Cloud
Secure your apps and infrastructure
Go to Microsoft Defender for Cloud >

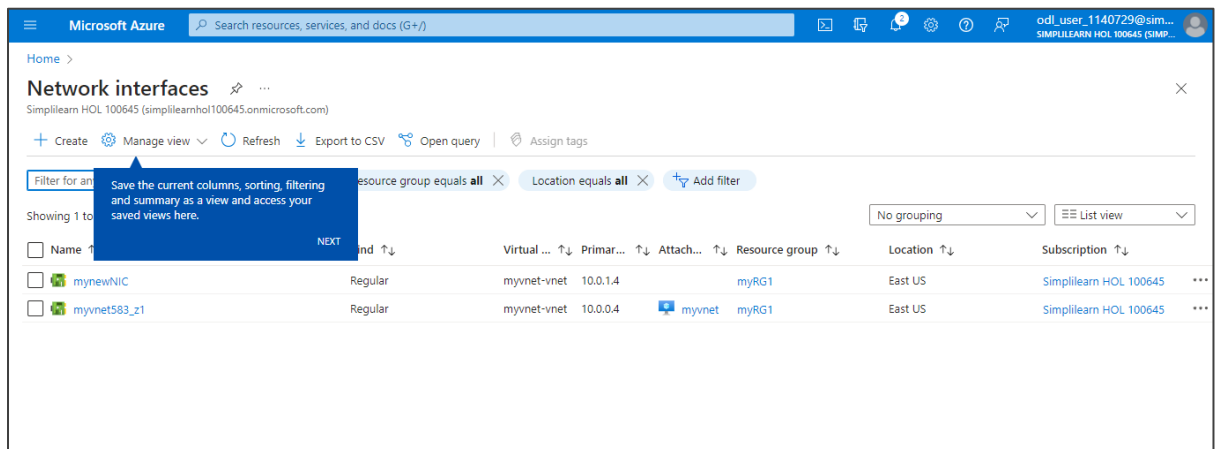
Free Microsoft tutorials
Start learning today >

Step 4: Create a network interface

4.1 In the Azure portal, search for and select **Network interfaces**



4.2 In the **Network interfaces** page, click on **+ Create**



4.3 In the **Create network interface** page, navigate to the **Basics** tab and enter the following details as shown below. Then, click on **Review + create**.

Microsoft Azure Search resources, services, and docs (G+)

Home > Network interfaces >

Create network interface

Basics Tags Review + create

Create a network interface and attach it to a virtual machine. A network interface enables a virtual machine to communicate with Internet, Azure, and on-premises resources. [Learn more about network interface](#)

Project details

Subscription *

Resource group * [Create new](#)

Instance details

Name * ✓

Region *

Virtual network * [Edit virtual network](#)

Subnet * [Edit subnet](#) 10.1.0.0 - 10.1.0.255 (256 addresses)

Microsoft Azure Search resources, services, and docs (G+)

Home > Network interfaces >

Create network interface

Resource group [Create new](#)

Instance details

Name * ✓

Region *

Virtual network * [Edit virtual network](#)

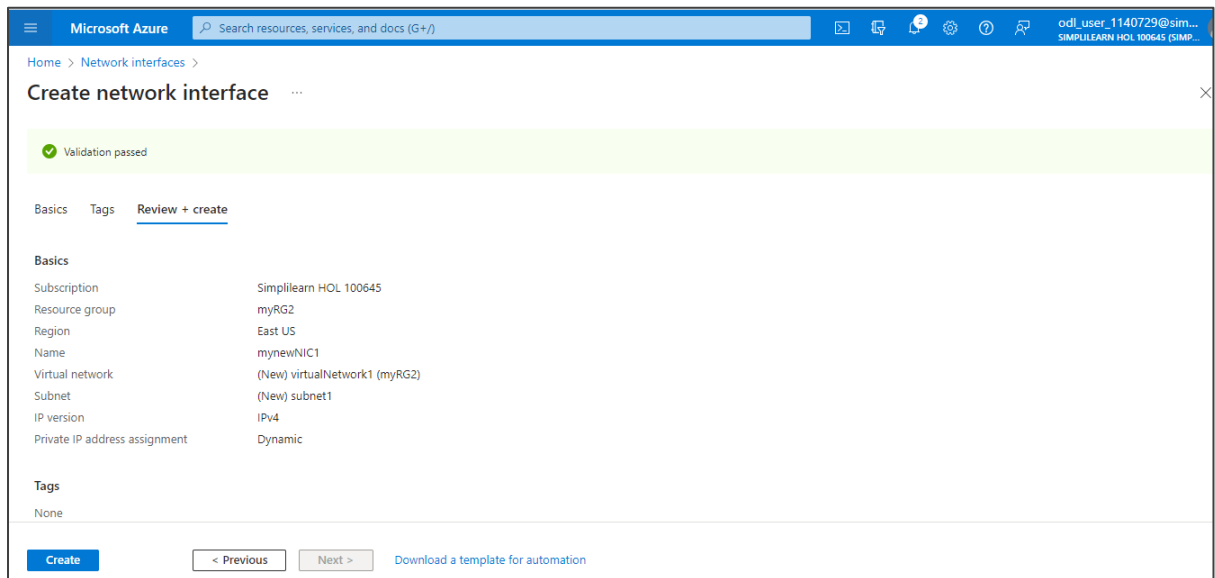
Subnet * [Edit subnet](#) 10.1.0.0 - 10.1.0.255 (256 addresses)

IP version ☒ IPv4 ☐ IPv4 and IPv6

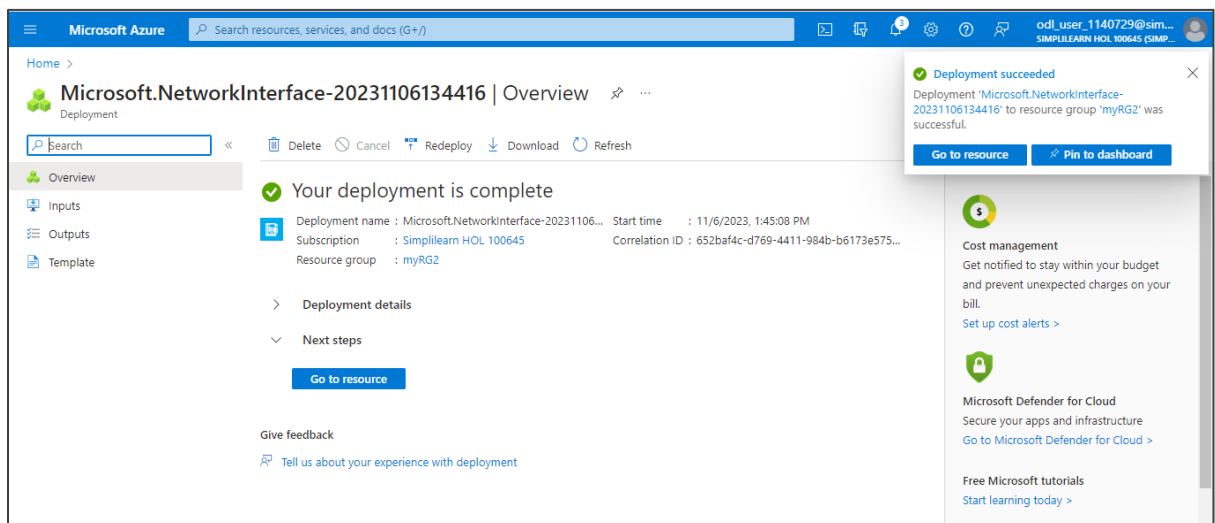
Private IP address assignment ☒ Dynamic ☐ Static

[Review + create](#) [Previous](#) [Next: Tags](#) [Download a template for automation](#)

4.4 After the validation passes, click on + Create

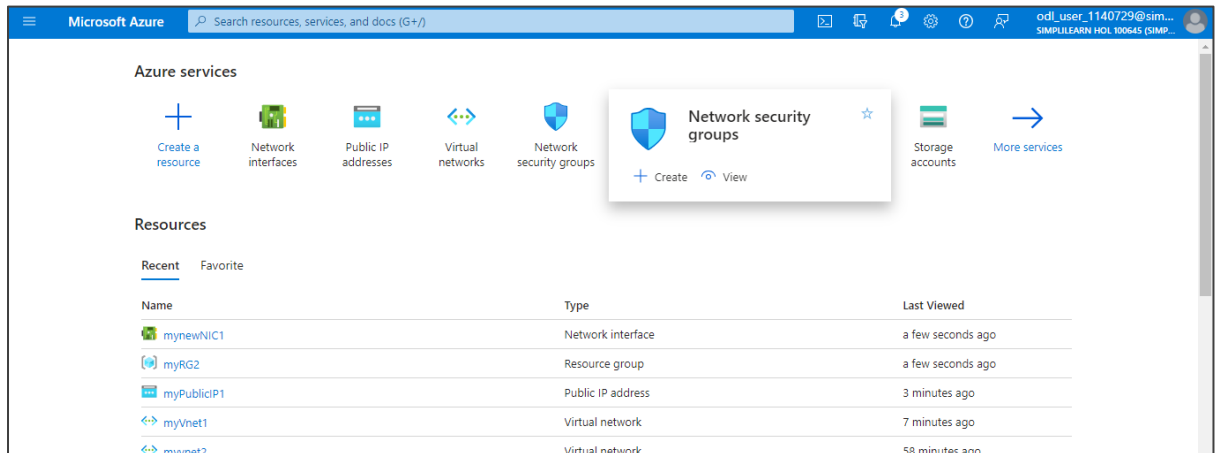


The network interface is created as shown below:

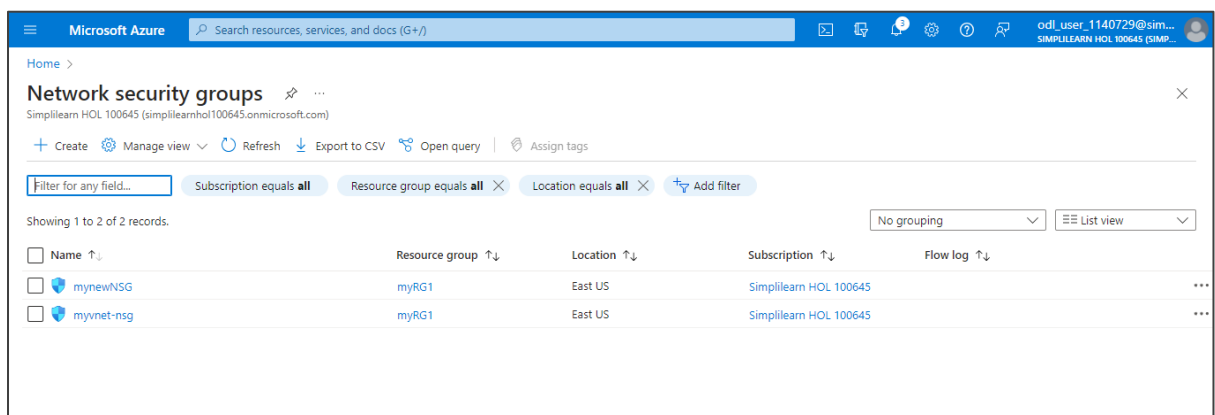


Step 5: Create a network security group

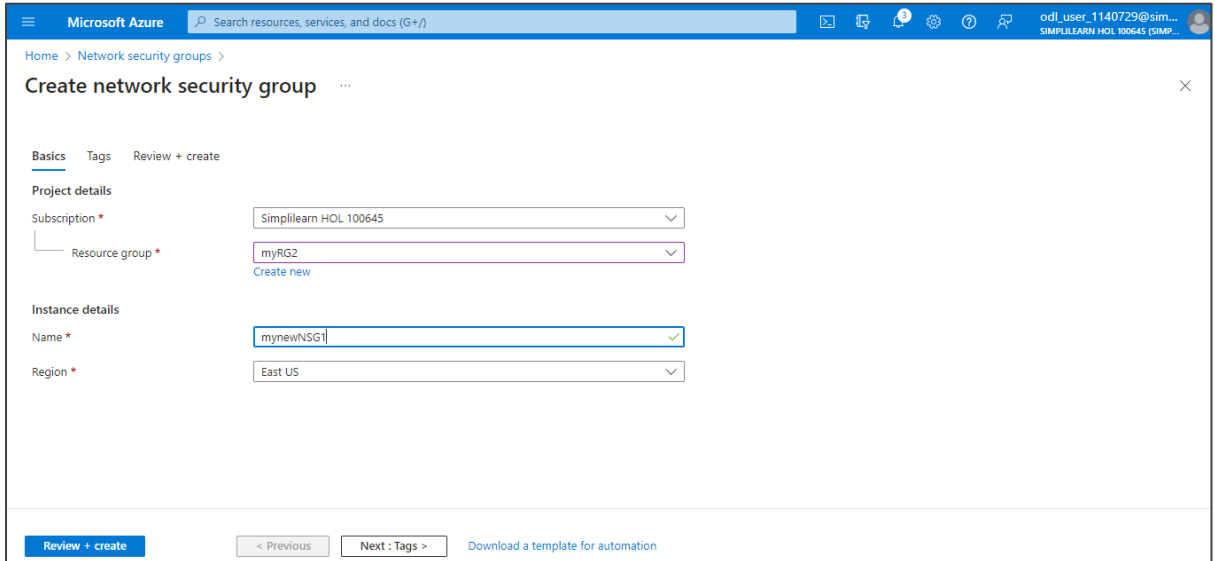
5.1 In the Azure portal, search for and select **Network security groups**



5.2 In the **Network security groups** page, click on **+ Create**



5.3 In the **Create network security group** page, navigate to the **Basics** tab and enter the details as shown below. Then, click on **Review + create**.



The screenshot shows the 'Create network security group' page in the Microsoft Azure portal. The 'Basics' tab is selected, and the 'Review + create' button is highlighted. The form contains the following details:

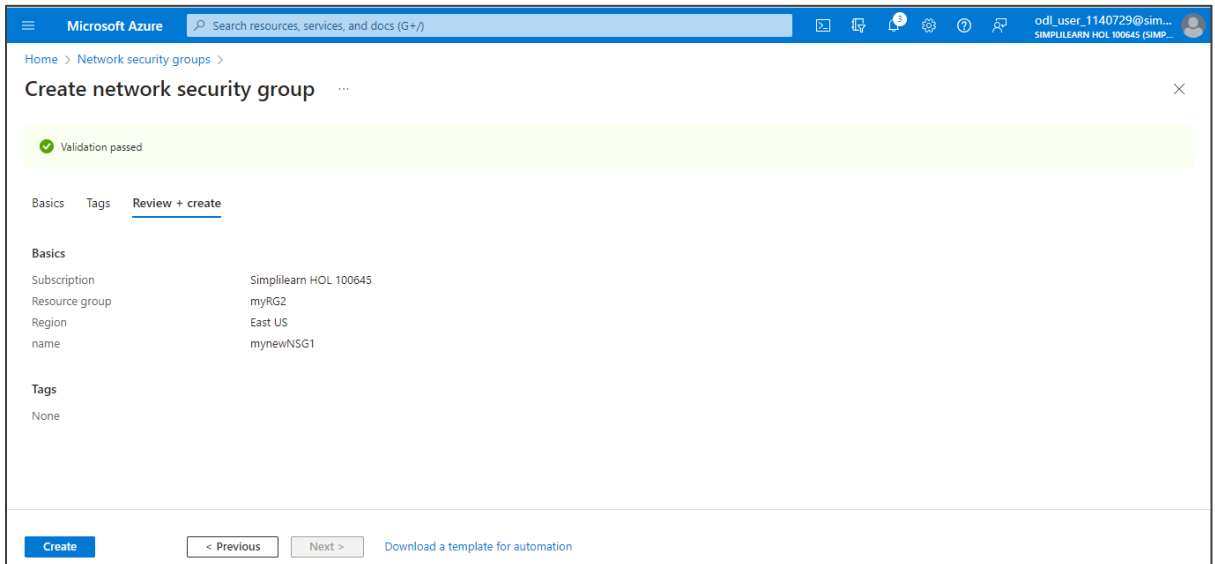
Project details	
Subscription *	Simplilearn HOL 100645
Resource group *	myRG2

[Create new](#)

Instance details	
Name *	mynewNSG1
Region *	East US

At the bottom, there are navigation buttons: **Review + create**, [< Previous](#), [Next : Tags >](#), and [Download a template for automation](#).

5.4 After the validation passes, click on **Create**



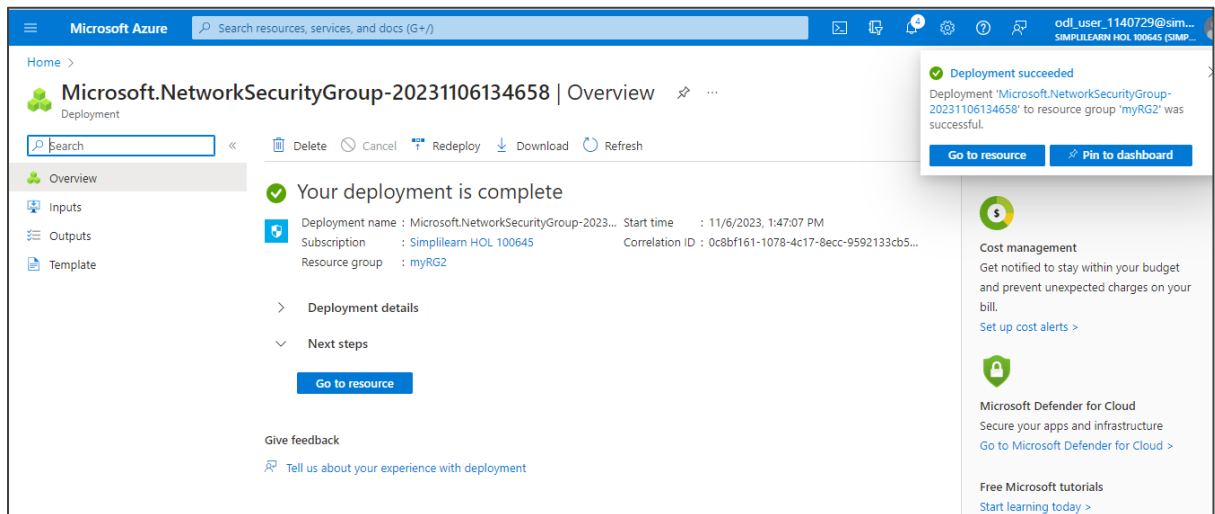
The screenshot shows the 'Create network security group' page in the Microsoft Azure portal, now at the 'Review + create' tab. A green banner at the top indicates 'Validation passed'. The 'Basics' tab is selected, and the 'Create' button is highlighted. The form displays the following details:

Basics	
Subscription	Simplilearn HOL 100645
Resource group	myRG2
Region	East US
name	mynewNSG1

Tags	
	None

At the bottom, there are navigation buttons: **Create**, [< Previous](#), [Next >](#), and [Download a template for automation](#).

The network security group is created as shown below:



By following these steps, you have successfully created Azure's virtual networking capabilities by creating a virtual network for Azure virtual machines, implementing network-based segmentation with static IP addresses, securing public endpoints, and enabling DNS name resolution within the virtual network.